

Smart Walker

Presented by VISION:

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The problem

"36 million falls are reported among older adults in the U.S. every year" —CDC

- Standard walkers provide physical support but no awareness of the environment — they can't warn users about obstacles, steps, or hazards ahead
- Elderly users often live alone and cannot rely on someone always being nearby to guide them safely from room to room



The mission.

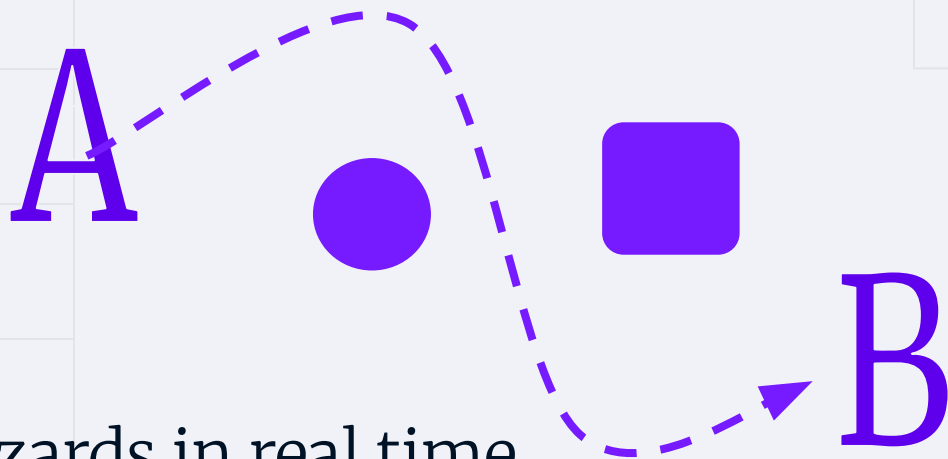
Empower people with visual impairments to live more independently through accessible AI technology



Smart Walker Overview

- ❖ Helps visually impaired elderly users navigate safely at home
- ❖ Detects obstacles using cameras and sensors to provides real-time voice alerts
- ❖ Guides users step-by-step to any location in their home through mapping software

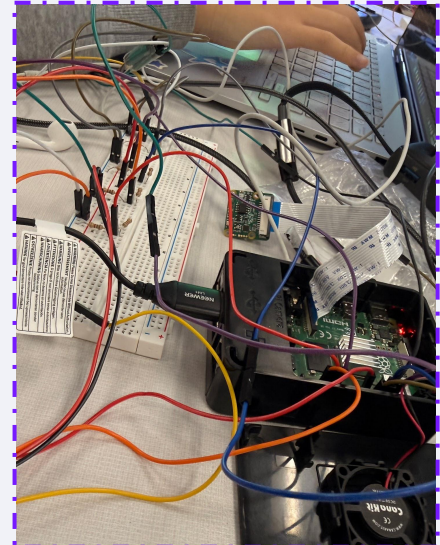
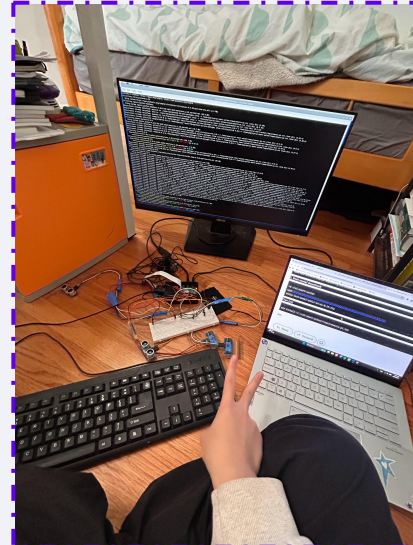
Key Functions



1. Detects obstacles + hazards in real time
2. Guides users from room to room with voice directions
3. Listens to and responds with audio input+output
4. Speaks alerts and instructions through a built-in speaker

Hardware overview

1. Raspberry Pi
2. Ultrasonic Sensors
3. 12 MP Pi Camera
4. USB Microphone
5. Speaker (Stereo Jack)
6. Breadboard
7. Rainbow Wires+Resistors
8. Walker
9. Ropes+Ribbons+Container



Design Flow

1. Walker powers on and greets the user
2. Asks the user where they currently are
3. User states their location by voice
4. User states their destination by voice
5. System calculates a route using a pre-mapped layout of the home
6. Walker speaks step-by-step directions to the destination
7. Along the way, the camera and sensors continuously detect obstacles and warn the user in real time
8. Walker announces when the destination has been reached



How we built it



Hardware Setup and Testing	User Input Setup	LLM integration	Navigation System
- Assembled Raspberry Pi with camera, ultrasonic sensors, microphone, and speaker - Tested each component individually	- Integrated offline speech recognition to capture voice commands from the user in real time - VOSK	- Connected AI model to generate natural, context-aware spoken alerts for obstacle warnings and navigation guidance - General output coded for offline use (local model)	- Built room-to-room route planning using a pre-mapped home layout - PyAudio: Delivers step-by-step voice directions with real-time obstacle awareness

Sample output for navigation

From Bathroom to Kitchen...

User: I am in the bathroom, I want to go to the kitchen.

Walker: You are currently in the bathroom, navigating to the kitchen. 3 steps total. Go out through the door to the hallway.

Walker: (door classifier triggers separately) The door ahead is closed, please open it to continue.

User: Continue

Walker: Walk to the end of the hallway and go through the door into the living room. 2 steps remaining.

Walker: (door classifier) The door ahead is closed, please open it to continue.



Roadblocks

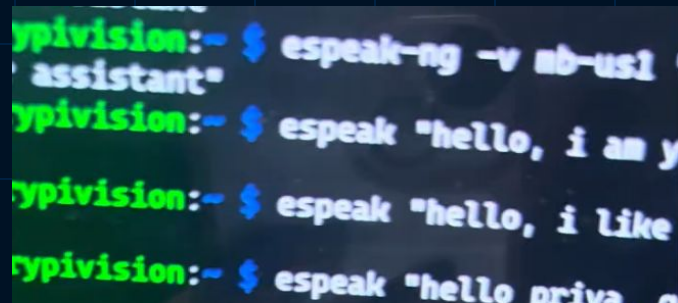
App to On-Device Processing

We transitioned from an app-based system to a fully internal Raspberry Pi processing system.

Processing speed of Raspberry Pi

We faced limitations in processing speed when running multiple systems simultaneously on the device.

Future Improvements



```
ypivision:~ $ espeak-ng -v nb-us1  
assistant"  
ypivision:~ $ espeak "hello, i am y  
ypivision:~ $ espeak "hello, i like  
ypivision:~ $ espeak "hello drive
```

- **Outdoor navigation** — extend the system beyond pre-mapped indoor layouts
- **Automatic map generation** — eliminate manual setup by having the walker learn the home on its own
- **Fall detection** — use the camera to detect falls and alert a caregiver
- **Faster processing** — upgrade hardware to improve real-time detection beyond current rate

Project Artifacts.

- **Github** — <https://github.com/priyaprabhudgp/ai-smart-walker>

- **Documentation** —

[https://drive.google.com/drive/folders/1z78CIVQ6NRQ4izaFbOxU4GfkqnKC1GNC?usp=drive link](https://drive.google.com/drive/folders/1z78CIVQ6NRQ4izaFbOxU4GfkqnKC1GNC?usp=drive_link)

- **Budget** —

https://docs.google.com/spreadsheets/d/1-1TdF_PpTounRCCMZyCLOmeC4ceP3yk1BXia3_T_Ijc/edit?gid=0#gid=0

- **Video Demo** — <https://drive.google.com/file/d/1Kr7q-EWURKGRKXhBwtN3RUdj6aManCDA/view?usp=sharing>

Thank You!

